

RANDOM WALK AND THE BOUNDARY VALUE PROBLEMS

BSc Thesis

written by

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Abstract

We commence by constructing simple random walk models in both one and higher dimensions, determining their probability distributions and examining their recurrent or transient properties. Subsequently, we apply these models to boundary value problems and utilise this approach to tackle the Dirichlet problem, a significant boundary value problem.

Keywords: random walk, probability, recurrence, transience, boundary value problems, Dirichlet problem.

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Chapter 1

Introduction

The random walk model, which initially piqued my interest through my financial mathematics curriculum, has captivated my imagination due to its versatile applicability across a multitude of disciplines, such as economics, physics, chemistry, and biology. One of the primary applications of random walks lies in modelling stock prices, which serves as the foundation for the renowned *Black-Scholes model*. The universality and practicality of the random walk model in tackling intricate problems are truly mesmerising, prompting me to delve deeper into its construction and properties. I am particularly intrigued by the mathematical applications of the random walk model. Consequently, I investigate the intricate connection between random walks and boundary value problems, and their potential in solving the *Dirichlet problem*, a specialised boundary value problem for harmonic functions.

In this report, we undertake this investigation by examining the one-dimensional simple random walk, delving into its probability distribution and ever-returning property. Subsequently, we extend the discussion to higher-dimensional cases. This then leads to a remarkable mathematical application of the random walk model for solving the *Dirichlet problem*. To draw the desired conclusions, we have also incorporated *Appendix 2*, which covers some prerequisite knowledge. This includes fundamental lattice definitions, the proof of *Stirling's formula*, and essential statistics and probability concepts. To present some of my conclusions more vividly, we have also provided Python examples in *Appendix A*.

The work of this report is primarily based on [Law10, Sections 1.1, 1.2, and 1.4]. I perused [Nor98] for additional inspiration related to recurrence and transience. In solving boundary value problems with the random walk models, I also referred to [Gua11]. For the Appendix, I consulted [HPS08, Chapter 7] for lattice concepts. In deriving *Stirling's formula*, I referred to [LS15, Section 15.2] and, specifically, [Abb01] and notes were written by A. Atminas, which were used for Analysis 2 during semester 2, 2020/2021, for Taylor's formula.

Literature Review

The random walk model boasts a rich history dating back to the early 20th century, with its origins traced to the work of Karl Pearson, a renowned British statistician. In 1905, Pearson laid the foundation for the random walk model by proposing a problem involving a walker taking n successive equal steps in a two-dimensional plane, with an equal probability of moving in all directions [Pea05]. The crux of the problem was to determine the walker's distance from the starting point after completing these steps. In the very same year, Lord Rayleigh, a distinguished British mathematician and physicist, derived the probability distribution for large enough values of n [Ray05]. Rayleigh's ingenuity also led

him to generalise the problem and obtain the solution for the three-dimensional case [Ray19].

In 1921, George Pólya, a brilliant Hungarian mathematician, continued research on this model. His original article [Pól21] was written in German, so we also refer to the English-language exposition in [Mar13]. Pólya’s groundbreaking work revealed that a simple random walk exhibits recurrent behaviour for one- and two-dimensional cases, while transient behaviour for higher-dimensional cases. This fascinating finding was later summarised by Shizuo Kakutani, a Japanese-American mathematician, who stated that “a drunk man will find his way home, but a drunk bird may get lost forever” [Dur19].

As the passage of time unfolded, the concept of random walks continued to evolve and expand. H. N. V. Temperley astutely observed that the problem could be extended to using lattices, paving the way for broader applications in various physical systems and models [Tem56]. Consequently, random walks found relevance in areas such as diffusion, Brownian motion, and the heat equation, profoundly impacting various branches of mathematics and physical sciences [Car77].

In summary, the study of random walks has come a long way since its inception. With its applications ranging from seemingly simple pedestrian movement to complex physical phenomena, random walks have been shown to be a versatile and powerful tool in mathematics and other science disciplines.

Chapter 2

Mathematical Backgrounds

2.1 Fundamentals of Lattice

A lattice is a captivating array of points in an n -dimensional space, composed of points displaying a periodic pattern, resembling a vector space. However, unlike a typical vector space, the vectors in a lattice can only be multiplied by integers, introducing a distinctive constraint to its structure. This section introduces the basic definitions related to lattices and provides an example to illustrate the concept.

Definition 2.1 (Lattice). Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be a set of linearly independent vectors. The *lattice* $\mathcal{L}$ generated by $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ is the set of linear combinations of these vectors with coefficients in $\mathbb{Z}$,

$$\mathcal{L} = \{a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n \mid a_1, a_2, \dots, a_n \in \mathbb{Z}\}.$$

A *basis* for $\mathcal{L}$ is any set of independent vectors that generates $\mathcal{L}$. The dimension of $\mathcal{L}$ is the number of vectors in a basis for $\mathcal{L}$.

Definition 2.2 (Integral Lattice). An *integral* (or *integer*) lattice is a lattice in which all vectors have integer coordinates.

Integral lattices are particularly appealing to work with, as the vectors possess integer coordinates, which aligns with the methodology discussed in this report.

Example 2.3. Consider the lattice

$$\mathbb{Z}^n = \{(x_1, x_2, \dots, x_n) \mid x_1, x_2, \dots, x_n \in \mathbb{Z}\},$$

which comprises all vectors with integer coordinates. The basis for this lattice is formed by an $n \times n$ matrix whose columns are the n standard basis vectors

$$\begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix},$$

resulting in the matrix

$$\begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

In the example above, $\mathbb{Z}^n$ represents an integral lattice, as every vector within the lattice has integer coordinates. The basis for this lattice is simply the identity matrix, which consists of the standard basis vectors for an n -dimensional space.

2.2 Probability and Statistics

In this section, we present some fundamental concepts of probability and statistics that underpin the study of random walks.

Definition 2.4 (Probability Space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ consisting of a non-empty sample space Ω , a σ -algebra $\mathcal{F}$ on Ω , and a probability measure $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$.

The sample space Ω represents the set of all possible outcomes of a given experiment. The σ -algebra $\mathcal{F}$ is a collection of subsets of Ω that are considered as “measurable” events, and it must satisfy certain properties. Finally, the probability measure $\mathbb{P}$ assigns a probability value to each event in $\mathcal{F}$, subject to a set of axioms, such as the probability of the entire sample space being equal to 1.

Definition 2.5 (Random Variable). A *random variable* X is a measurable function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to a measurable space $(S, \mathcal{S})$.

In other words, a random variable maps each outcome in the sample space to a value in S in such a way that the pre-image of every measurable set in $\mathcal{S}$ belongs to $\mathcal{F}$.

Remark. *Random variables are generally not written explicitly as functions on the state space. For a measurable set $E \in \mathcal{S}$, for example, we write*

$$\mathbb{P}(X^{-1}(E)) = \mathbb{P}(X \in E).$$

This notation captures the idea that we are interested in the probability of the event where the random variable X takes a value in the set E .

Proposition 2.6 (Countable Additive Rule). *If $E_1, E_2, \dots$ are disjoint (mutually exclusive) events, then*

$$\mathbb{P}\left(\bigcup_{n=1}^{\infty} E_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(E_n).$$

The Countable Additive Rule is one of the fundamental properties of a probability measure, which is essential for working with an infinite number of events, and it ensures that the probability measure maintains its consistency and coherency when dealing with countably infinite sets.

Definition 2.7 (Independence). Events E_1 and E_2 are *independent* if

$$\mathbb{P}(E_1 \cap E_2) = \mathbb{P}(E_1)\mathbb{P}(E_2).$$

Random Variables X_1 and X_2 in $\mathbb{R}$ are *independent* if for all $x_1, x_2 \in \mathbb{R}$, we have

$$\mathbb{P}\{(X_1 \geq x_1) \cap (X_2 \geq x_2)\} = \mathbb{P}(X_1 \geq x_1)\mathbb{P}(X_2 \geq x_2).$$

Definition 2.8 (Conditional Probability). For events E_1 and E_2 with $\mathbb{P}(E_2) > 0$, the *conditional probability* of E_1 occurring given that E_2 has occurred is

$$\mathbb{P}(E_1 | E_2) = \frac{\mathbb{P}(E_1 \cap E_2)}{\mathbb{P}(E_2)}.$$

The independence of random variables implies that the behaviour of one does not provide any information about the other. The conditional probability, on the other hand, measures the likelihood of E_1 happening when we have the information that E_2 has already taken place.

Definition 2.9 (Expectation). For discrete random variables, the *expectation* (or *expected value*, *mean*, *average value*) $\mathbb{E}$ is denoted as

$$\mathbb{E}[X] = \sum_z z\mathbb{P}(X = z).$$

Definition 2.10 (Variance). For discrete random variables, the *variance* Var is denoted as

$$\begin{aligned} \text{Var}[X] &= \mathbb{E}[(X - \mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] \\ &= \mathbb{E}[X^2] - 2\mathbb{E}[X]\mathbb{E}[X] + (\mathbb{E}[X])^2 = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \end{aligned}$$

Expectation acts as a measure of central tendency or average value of a random variable, encapsulating its most typical outcome. Variance, on the other hand, provides a measure of the dispersion or spread of the values around the mean, indicating the degree of randomness or uncertainty in the data.

Proposition 2.11. For random variables $X_1, \dots, X_n$ with finite expectations,

$$\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n].$$

If the random variables are also independent and have finite variances, then

$$\text{Var}[X_1 + \dots + X_n] = \text{Var}[X_1] + \dots + \text{Var}[X_n].$$

2.3 Deriving Stirling's Formula

In this section, we derive *Stirling's formula* using *Wallis' formula* and a lemma about monotone functions.

Lemma 2.12.

$$\int_0^\pi \sin^{2n} x dx = \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots 2n} \pi$$

where $n \in \mathbb{N}^*$, and

$$\int_0^\pi \sin^{2n+1} x dx = \frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n+1)} \cdot 2$$

where $n \in \mathbb{N}$.

Proof. Let $I_k = \int_0^\pi \sin^k x dx$ for all $k \in \mathbb{N}$. Then

$$\begin{aligned} I_{k+1} &= \int_0^\pi \sin^{k+1} x dx \\ &= \int_0^\pi \sin^2 x \cdot \sin^{k-1} x dx \\ &= \int_0^\pi (1 - \cos^2 x) \cdot \sin^{k-1} x dx \\ &= \int_0^\pi \sin^{k-1} x dx - \int_0^\pi (\cos^2 x \cdot \sin^{k-1} x) dx = I_{k-1} - \int_0^\pi \cos x \cdot (\sin^{k-1} x \cdot \cos x) dx. \end{aligned}$$

By integration by parts,

$$\begin{aligned} \int_0^\pi \cos x \cdot (\sin^{k-1} x \cdot \cos x) dx &= \int_0^\pi \cos x \cdot \left(\frac{1}{k} \cdot \sin^k x \right)' dx \\ &= \left[\cos x \cdot \frac{1}{k} \cdot \sin^k x \right]_0^\pi - \int_0^\pi \frac{1}{k} \cdot \sin^k x \cdot (-\sin x) dx = \frac{1}{k} \cdot I_{k+1}. \end{aligned}$$

Therefore, we have

$$I_{k+1} = I_{k-1} - \frac{1}{k} I_{k+1},$$

and hence $I_{k+1} = \frac{k}{k+1} I_{k-1}$. Consequently,

$$I_{2n} = \frac{2n-1}{2n} I_{2n-2} = \cdots = \frac{2n-1}{2n} \cdot \frac{2n-3}{2n-2} \cdots \frac{1}{2} \cdot I_0,$$

which exactly matches the first equation, and the second equation follows similarly. $\square$

Theorem 2.13 (Wallis' Formula).

$$\pi = \lim_{n \rightarrow \infty} \left[\frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n-1)} \right]^2 \cdot \frac{1}{n}.$$

Proof. Given that $\sin^{2n-1} x \geq \sin^{2n} x \geq \sin^{2n+1} x$ for all $x \in [0, \pi]$, it follows that $I_{2n-1} \geq I_{2n} \geq I_{2n+1}$. Consequently, we arrive at the following inequality:

$$\frac{2 \cdot 4 \cdots (2n-2)}{1 \cdot 3 \cdots (2n-1)} \cdot 2 \geq \frac{1 \cdot 3 \cdots (2n-1)}{2 \cdot 4 \cdots 2n} \pi \geq \frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n+1)} \cdot 2,$$

which yields

$$\left[\frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n-1)} \right]^2 \cdot \frac{1}{n} \geq \pi \geq \left[\frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n-1)} \right]^2 \cdot \frac{2}{2n+1}.$$

Now, let us define the sequence W_n as

$$W_n = \left[\frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n-1)} \right]^2 \cdot \frac{1}{n},$$

which enables us to rewrite the previous inequality as

$$W_n \geq \pi \geq W_n \cdot \frac{2n}{2n+1},$$

and we obtain

$$\pi \leq W_n \leq \pi \cdot \frac{2n+1}{2n}.$$

By applying the squeeze theorem, we deduce that $\lim_{n \rightarrow \infty} W_n = \pi$, thereby establishing *Theorem 2.13*. $\square$

Lemma 2.14. *Let f and g be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = g(a)$ and $a < x \leq b$ implies $f'(x) \geq g'(x)$, then $f(x) \geq g(x)$ for all $x \in [a, b]$. Also, if $f(a) = g(a)$ and $a < x \leq b$ imply $f'(x) > g'(x)$, then $f(x) > g(x)$ for all $x \in (a, b]$.*

Proof. Let $h = f - g$. If $a < x \leq b$ implies $f'(x) \geq g'(x)$, then $h'(x) \geq 0$ for all $x \in [a, b]$, and so h is monotone increasing on $[a, b]$. If $f(a) = g(a)$, then $h(a) = 0$, and so $h(x) \geq h(a) = 0$, and thus $f(x) \geq g(x)$ for all $x \in [a, b]$. The second statement follows similarly. $\square$

Example 2.15. Prove that

$$\log(1+x) > \frac{2x}{x+2}$$

for all $x > 0$.

Proof. By invoking *Lemma 2.14*, it is sufficient to establish the inequality between the derivatives of the left- and right-hand sides. Our goal, therefore, is to show that

$$\frac{1}{x+1} > \frac{4}{(x+2)^2}$$

for all $x > 0$. Multiplying both sides by $(x+1)(x+2)^2$ gives $(x+2)^2 > 4(x+1)$, which simplifies to $x^2 > 0$. Consequently, the inequality holds for all $x > 0$. $\square$

Lemma 2.16 (Mean Value Theorem). *Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists a point $c \in (a, b)$ such that*

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Theorem 2.17 (Taylor's Theorem). *Let $f \in C([a, b])$, and assume that the $(n+1)$ -th derivative $f^{(n+1)}$ exists for all $x \in (a, b)$. Then, for any $c, x \in (a, b)$, there exists a point y lying between c and x such that*

$$f(x) = f(c) + f'(c)(x-c) + \cdots + \frac{f^{(n)}(c)}{n!}(x-c)^n + \frac{f^{(n+1)}(y)}{(n+1)!}(x-c)^{n+1}.$$

Proof. Fix a point $c \in (a, b)$, and define the polynomial

$$P(t) = f(c) + f'(c)(t-c) + \cdots + \frac{f^{(n)}(c)}{n!}(t-c)^n.$$

Choose a point $x \in (a, b)$ with $x \neq c$ and define

$$M = \frac{f(x) - P(x)}{(x-c)^{n+1}} \in \mathbb{R}.$$

Consider the function

$$g(t) = f(t) - P(t) - M(t-c)^{n+1}.$$

By the definition of M , we have $g(x) = 0$. Thus,

$$f(x) = P(x) + M(x-c)^{n+1},$$

and our goal is to show that there exists some point y between x and c such that $M = \frac{f^{(n+1)}(y)}{(n+1)!}$. We will establish this by considering the derivatives of $g(t)$ and applying *Lemma 2.16*.

Differentiating $g(t)$ term by term and evaluating at c , we find that

$$g(c) = g'(c) = \cdots = g^{(n)}(c) = 0.$$

Since $g(x) = g(c) = 0$, *Lemma 2.16* implies that $g'(x_1) = 0$ for some point x_1 lying between x and c . Now, as $g'(x_1) = g'(c)$, there must exist some point x_2 between x_1 and c such that $g''(x_2) = 0$. By induction, there exists a point x_{n+1} such that $g^{(n+1)}(x_{n+1}) = 0$, where $y := x_{n+1}$ lies between x and c . Therefore, we have $g^{(n+1)}(y) = 0$. Moreover, by differentiating $g(t)$ term by term and evaluating at y , we obtain

$$g^{(n+1)}(y) = f^{(n+1)}(y) - (n+1)!M.$$

Consequently, the right-hand side equals zero, which implies that $M = \frac{f^{(n+1)}(y)}{(n+1)!}$. Thus, we have demonstrated the existence of a point y between x and c satisfying the desired property. $\square$

Example 2.18. The Taylor series expansion of $\log(1+x)$ centred at 0 is given by

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots,$$

and we can show that

$$\log(1+x) \leq x - \frac{x^2}{2} + \cdots + \frac{x^{2n+1}}{2n+1}$$

for all $x > 0$, where $n \in \mathbb{N}$.

Proof. Consider the function

$$f(x) = x - \frac{x^2}{2} + \cdots + \frac{x^{2n+1}}{2n+1} - \log(1+x),$$

and our goal is to show that $f(x) \geq 0$ for all $x > 0$. To do this, let us first examine the derivative

$$f'(x) = 1 - x + \cdots + x^{2n} - \frac{1}{1+x} = -\frac{(-x)^{2n+1}}{1+x} > 0$$

for all $x > 0$. This inequality implies that $f(x)$ is an increasing function for $x > 0$. In addition, we note that $f(0) = 0$. Consequently, we can conclude that $f(x) \geq 0$ for all $x > 0$. $\square$

Proof of Theorem 3.1. We start by examining the sequence

$$a_n = \frac{\sqrt{2\pi n}^{n+\frac{1}{2}} e^{-n}}{n!},$$

to demonstrate that it is strictly monotone increasing and bounded. To achieve this, we compute the ratio of consecutive terms

$$\frac{a_{n+1}}{a_n} = \frac{(1 + \frac{1}{n})^{n+\frac{1}{2}}}{e}.$$

Consequently, we obtain the following expression

$$\log a_{n+1} - \log a_n = \left(n + \frac{1}{2}\right) \log \left(1 + \frac{1}{n}\right) - 1. \quad (2.1)$$

for all n . Recalling *Example 2.15*, we know that

$$\log(1+x) > \frac{2x}{x+2}$$

for $x > 0$. Substituting $x = \frac{1}{n}$ and multiplying both sides by $n + \frac{1}{2}$, we deduce

$$\left(n + \frac{1}{2}\right) \log\left(1 + \frac{1}{n}\right) > 1.$$

This result confirms that the sequence $(a_n)_n$ is strictly monotone increasing.

Drawing upon *Example 2.18*, we have the inequality

$$\log(1+x) \leq x - \frac{x^2}{2} + \frac{x^3}{3}.$$

To find an upper bound for the right-hand side of (2.1), we substitute $x = \frac{1}{n}$ and multiply through by $n + \frac{1}{2}$. This gives us

$$\begin{aligned} \log a_{n+1} - \log a_n &\leq \frac{1}{12n^2} + \frac{1}{6n^3} \\ &\leq \frac{1}{12n^2} + \frac{1}{6n^2} = \frac{1}{4n^2}. \end{aligned}$$

Thus,

$$\begin{aligned} \log a_n - \log a_1 &= \sum_{i=1}^{n-1} (\log a_{i+1} - \log a_i) \\ &\leq \sum_{i=1}^{n-1} \frac{1}{4i^2} < \frac{1}{2} \end{aligned}$$

for all n . This analysis clearly shows that the sequence $(a_n)_n$ is bounded. Since we have demonstrated that $(a_n)_n$ is monotone increasing and bounded, we can conclude that it must be convergent. Since

$$\frac{2 \cdot 4 \cdots 2n}{1 \cdot 3 \cdots (2n-1)} = \frac{(2 \cdot 4 \cdots 2n)^2}{1 \cdot 2 \cdots (2n)} = \frac{(2^n \cdot n!)^2}{(2n)!} = \frac{4^n}{\binom{2n}{n}},$$

by *Theorem 2.13*, we have

$$\lim_{n \rightarrow \infty} \frac{4^n}{\binom{2n}{n} \sqrt{n}} = \sqrt{\pi}.$$

If $a_n \rightarrow L$, then

$$L = \lim_{n \rightarrow \infty} \frac{a_n^2}{a_{2n}} = \lim_{n \rightarrow \infty} \frac{\binom{2n}{n} \sqrt{\pi n}}{4^n} = 1.$$

□

Chapter 3

Simple Random Walk Model

We commence our exploration by delving into one of the most fundamental random walk models: the *simple random walk on the integer lattice $\mathbb{Z}^d$* .¹ In this model, the random walker advances one unit of length in either direction along the lattice for each unit of time that elapses. To better understand this model, we shall first examine the relatively straightforward one-dimensional case before venturing into the more intricate higher-dimensional scenarios.

3.1 One-Dimensional Random Walks

Imagine a one-dimensional world, represented by a number line upon which our random walker is initially placed. We take the starting point of the random walker to be the origin for convenience; however, the starting point can be chosen arbitrarily. With each passing moment, the random walker has an equal probability of moving one unit length to the left or right, as if guided by the whims of a fair coin toss dictating their every step.

Let us denote the *position* of the walker at time n by S_n . We can express S_n as the sum of the individual steps the walker takes, that is

$$S_n = X_1 + \cdots + X_n,$$

where $X_i = \pm 1$ are *independent and identically distributed (i.i.d.)* random variables denoting the change in position between time $i - 1$ and time i , with each step taken having equal probability $\mathbb{P}(X_i = 1) = \mathbb{P}(X_i = -1) = \frac{1}{2}$.

As the random walk progresses, the walker may find themselves back at the origin or venture far away from it, all depending on the accumulated outcomes of the coin tosses. Intrinsically, we yearn to know whether we can pinpoint the precise location of the walker at any given moment or, perhaps more generally, the probability distribution of the walker's position (*Section 3.2*). Moreover, we are fascinated by the question of whether the random walker will perpetually return to its starting point or eventually leave it behind forever (*Section 3.3*). The allure of this model stems from its simplicity, yet it furnishes a powerful foundation for unravelling the more elaborate higher-dimensional random walks (*Section 3.4*).

Now, let us investigate some of the basic properties of the model.² Since the random walker moves

¹Refer to *Section 2.1* for a description of the lattice.

²For a solid statistical background on calculations, kindly refer to *Section 2.2*.

left and right with equal probability, it is delightfully straightforward to find that $\mathbb{E}[S_n] = 0$. As we proceed to explore the variance, we unveil the following:

$$\begin{aligned}\mathbb{E}[S_n^2] &= \mathbb{E}\left[\left(\sum_{j=1}^n X_j\right)^2\right] \\ &= \mathbb{E}\left[\sum_{j=1}^n \sum_{k=1}^n X_j X_k\right] \\ &= \sum_{j=k} \mathbb{E}[X_j X_k] + \sum_{j \neq k} \mathbb{E}[X_j X_k] \\ &= \sum_{j=1}^n \mathbb{E}[X_j^2] + \sum_{j \neq k} \mathbb{E}[X_j X_k] = n + \sum_{j \neq k} \mathbb{E}[X_j X_k].\end{aligned}$$

Upon close observation, we notice that $\sum_{j \neq k} \mathbb{E}[X_j X_k] = 0$, as the probabilities of $X_j X_k = 1$ and $X_j X_k = -1$ are equal among the four random combinations of X_j and X_k being ± 1 . Thus, we find that $\text{Var}(S_n) = n$, which implies that the *typical distance* of the random walker from the origin is of order $\sqrt{n}$. This discovery signifies that after n steps, our random walker is expected to be approximately $\sqrt{n}$ steps away from their starting point.

Remark. *The vanishing of cross terms $\mathbb{E}[X_j X_k]$ is analogous to the perpendicularity or orthogonality of vectors in the Pythagorean theorem. This remarkable phenomenon arises from the fact that the random variables are independent and have a mean of zero, intertwining the realms of geometry and probability.*

3.2 Central Limit Theorem

As we delve deeper into this model, we find that determining the probability distribution of the random walk necessitates a greater level of effort. Consider a random walker who takes $2n$ steps and returns to their initial point. In this scenario, the walker must take n steps both to the left and to the right, as each move takes one unit of length. We employ the combinatorial notation $C(2n, n)$ to choose n of $2n$ steps as moves to the right. Consequently, the remaining n steps must be leftward, and the probability of this occurrence is 2^{-2n} . Thus, we have

$$\mathbb{P}(S_{2n} = 0) = \binom{2n}{n} 2^{-2n} = \frac{(2n)!}{n!n!} 2^{-2n}.$$

To explore more general scenarios, let us consider the probability of our random walker reaching a position of $2j$. In this case, we need to choose $n + j$ steps to the right and $n - j$ steps to the left. This yields the following expression:

$$\mathbb{P}(S_{2n} = 2j) = \binom{2n}{n+j} 2^{-2n} = \frac{(2n)!}{(n+j)!(n-j)!} 2^{-2n}. \quad (3.1)$$

Our next challenge is to unravel the meaning of this intriguing expression, which requires a deep understanding of the behaviour of $n!$ as n approaches infinity.

Theorem 3.1 (Stirling's Formula). *As $n \rightarrow \infty$,*

$$n! \sim \sqrt{2\pi n} n^{n+\frac{1}{2}} e^{-n},$$

where $\sim$ indicates that the ratio of the two sides converges to 1.

The proof is given in *Section 2.3*. *Theorem 3.1* enables us to derive an asymptotic expression for the probability distribution in (3.1). Applying the theorem to the expression and taking the limit as $n \rightarrow \infty$, we obtain

$$\begin{aligned} \mathbb{P}(S_{2n} = 2j) &\sim \frac{\sqrt{2\pi}(2n)^{2n+\frac{1}{2}}e^{-2n}}{\sqrt{2\pi}(n+j)^{n+j+\frac{1}{2}}e^{-(n+j)}\sqrt{2\pi}(n-j)^{n-j+\frac{1}{2}}e^{-(n-j)}} 2^{-2n} \\ &\sim \frac{1}{\sqrt{\pi}} \frac{n^{2n+\frac{1}{2}}(n+j)^{-n-j}(n-j)^{-n+j}}{(n+j)^{\frac{1}{2}}(n-j)^{\frac{1}{2}}} \\ &\sim \frac{1}{\sqrt{\pi}} \frac{n^{2n}(n^2-j^2)^{-n}(n+j)^{-j}(n-j)^jn^{\frac{1}{2}}}{(n^2-j^2)^{\frac{1}{2}}} \\ &\sim \frac{1}{\sqrt{\pi}} \frac{(n^2-j^2)^{-n}(n+j)^{-j}(n-j)^jn^{\frac{1}{2}}}{n^{-2n}n^{-j}n^j(n^2-j^2)^{\frac{1}{2}}} \\ &\sim \frac{1}{\sqrt{\pi}} \left(1 - \frac{j^2}{n^2}\right)^{-n} \left(1 + \frac{j}{n}\right)^{-j} \left(1 - \frac{j}{n}\right)^j \left(\frac{n}{n^2-j^2}\right)^{\frac{1}{2}}. \end{aligned} \quad (3.2)$$

Initially, let us consider the relatively simple case when $j = 0$, and we find that

$$\mathbb{P}(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi}} n^{-\frac{1}{2}}.$$

This result suggests that $\mathbb{P}(S_{2n} = 2j)$ also has a form of order $n^{-1/2}$. Let us now substitute $j = r\sqrt{n}$ into the right-hand side of (3.2). As a result, we obtain

$$\frac{1}{\sqrt{\pi n}} \left(1 - \frac{r^2}{n}\right)^{-n} \left[\left(1 + \frac{r}{\sqrt{n}}\right)^{\sqrt{n}}\right]^{-r} \left[\left(1 - \frac{r}{\sqrt{n}}\right)^{-\sqrt{n}}\right]^{-r} \left(\frac{1}{1 - \frac{r^2}{n}}\right)^{\frac{1}{2}}.$$

As n approaches infinity, this expression becomes asymptotic to

$$\frac{1}{\sqrt{\pi n}} e^{r^2} e^{-r^2} e^{-r^2} = \frac{1}{\sqrt{\pi n}} e^{-\frac{r^2}{n}}.$$

For every $a < b$,

$$\lim_{n \rightarrow \infty} \mathbb{P}\left(a\sqrt{2n} \leq S_{2n} \leq b\sqrt{2n}\right) = \lim_{n \rightarrow \infty} \sum \frac{1}{\sqrt{\pi n}} e^{-\frac{j^2}{n}}, \quad (3.3)$$

where the sum is taken over all j such that $a\sqrt{2n} \leq 2j \leq b\sqrt{2n}$. With $x_j = 2j/\sqrt{2n}$, the mesh size is $\sqrt{2/n}$, so the right-hand side is a Riemann sum approximation of an integral. This insight allows us to conclude that the limit converges to the integral

$$\int_a^b \frac{1}{C} e^{-\frac{x^2}{2}} dx,$$

where C is a constant. To ensure that the limiting distribution is indeed a probability distribution, we

must have

$$\int_{-\infty}^{\infty} \frac{1}{C} e^{-\frac{x^2}{2}} dx = 1.$$

By evaluating the integral,

$$\left(\int_{-\infty}^{\infty} e^{-\frac{x^2}{2}} dx \right)^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{x^2+y^2}{2}} dx dy = \int_0^{2\pi} \int_0^{\infty} e^{-\frac{r^2}{2}} r dr d\theta = 2\pi,$$

we find that $\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$. Consequently, the limit evaluates to

$$\int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

The convergence in (3.3) represents a manifestation of the renowned *Central Limit Theorem* (CLT) for the random walk model:

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(a\sqrt{2n} \leq S_{2n} \leq b\sqrt{2n} \right) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx.$$

Example 3.2. We can construct a one-dimensional random walk in Python and verify that it meets the *Central Limit Theorem*. To do this, we can simulate a large number of random walks, each consisting of multiple steps. By computing the final position of each walk, we can then compare the distribution of these positions to the Gaussian distribution predicted by the *Central Limit Theorem*. The implementation is given in *Appendix A*.

With this result, we have ventured further into the mysteries of the simple random walk, uncovering the behaviour of the probability distribution. One question that naturally arises is whether the random walker will ever return to its starting point, the origin, which we shall explore in the next section.

3.3 Returning to the Origin

We say a random walk is *recurrent* if it returns to the origin infinitely often with probability one, and *transient* otherwise. The following theorem establishes a crucial property of one-dimensional random walks.

Theorem 3.3. *A one-dimensional random walk is recurrent.*

Proof. In the realm of random walks, a fascinating aspect is the *Markov property*, which states that once a random walker reaches a certain state, its actions are independent of how it arrived at that state. This independence arises because the future state depends solely on the present state and the next step, which, in turn, hinges on independent and identically distributed random variables that are independent of the preceding step.

Given the Markov property, a one-dimensional random walk that returns to the origin seems to experience a form of amnesia, as it “forgets” its return and, thus, may revisit the origin. This intriguing behaviour motivates us to define a random walk that revisits the origin multiple times. Let us define $J_n = 1_{S_n=0}$. Here, we use the *indicator function* notation: if E is an event, then 1_E is the random variable that takes the value 1 if the event occurs and 0 if it does not occur. Let the total number of

visits to the origin by the random walker be

$$V = \sum_{n=0}^{\infty} J_{2n}.$$

The choice of even indices here is crucial, as it reflects the fact that a random walker, starting from the origin, can only return to the origin after an even number of steps. To explore this concept further, it is reasonable to calculate the expected value, $\mathbb{E}[V]$, since

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{E}[J_{2n}] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0).$$

Recalling the asymptotic expression

$$\mathbb{P}(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi}} n^{-\frac{1}{2}}$$

as $n \rightarrow \infty$, derived in the previous section, we find that

$$\mathbb{E}[V] = \infty.$$

This conclusion stems from the fact that the sum $\sum_{n=1}^{\infty} n^{-a}$ converges if $a > 1$ and diverges otherwise.

Example 3.4 (St. Petersburg Paradox). It is possible for a random variable to be finite yet have an infinite expectation. A well-known example of such a random variable arises from the St. Petersburg Paradox. Imagine playing a game where you flip a coin until you get tails. If you obtain k heads before flipping tails, your payoff is 2^k . The probability of getting exactly k heads is the probability of experiencing k consecutive heads followed by tails, which is $2^{-(k+1)}$. Consequently, the expected payoff in this game is

$$2^0 \cdot \frac{1}{2} + 2^1 \cdot \frac{1}{2^2} + 2^2 \cdot \frac{1}{2^3} + \cdots = \frac{1}{2} + \frac{1}{2} + \cdots + \frac{1}{2} = \infty.$$

Since the expectation is infinite, one should theoretically be willing to spend any amount of money to play this game once. However, this is not the case in reality, and herein lies the paradox.

Therefore, to demonstrate that V is indeed infinite, we must delve further into the analysis. Let q represent the probability that the random walker ever returns to the origin after time 0. We will show that $q = 1$ by initially assuming $q < 1$ and subsequently deriving a contradiction. Suppose $q < 1$. Under this assumption, we can derive the distribution for V . For instance, $\mathbb{P}(V = 1) = 1 - q$, as $V = 1$ if and only if the walker never returns after time zero. More generally, we have

$$\mathbb{P}(V = k) = q^{k-1}(1 - q),$$

where $k \in \mathbb{N}^*$. This relationship implies that

$$\mathbb{E}[V] = \sum_{k=1}^{\infty} k \mathbb{P}(V = k) = \sum_{k=1}^{\infty} k q^{k-1} (1 - q) = \frac{1}{1 - q} < \infty;$$

however, we have previously established that $\mathbb{E}[V] = \infty$. Thus, it must be the case that $q = 1$. Therefore, the probability that a one-dimensional simple random walker returns to the origin infinitely often is one, which implies recurrence. $\square$

Remark. Another way to compute $\mathbb{E}[V]$ in terms of the probability of returning to the origin q is to argue that

$$\mathbb{E}[V] = 1 + q\mathbb{E}[V],$$

where the 1 signifies the first visit to the origin. The crux of this argument lies in the observation that, given a second visit to the origin, the expected number of visits after the first visit is precisely equal to the expected number of visits starting at the origin. Solving this simple equation gives $\mathbb{E}[V] = (1 - q)^{-1}$.

3.4 Higher-dimensional Random Walks

We now consider a random walker on the d -dimensional integer lattice

$$\mathbb{Z}^d = \{(x_1, x_2, \dots, x_d) \mid x_1, x_2, \dots, x_d \in \mathbb{Z}\}.$$

The random walker chooses one of its $2d$ closest neighbours, each with probability $\frac{1}{2d}$, and moves to that point at each time step. Let

$$S_n = x + X_1 + \dots + X_n$$

denote the position of the particle where $x, X_1, \dots, X_n, S_n$ represent points in $\mathbb{Z}^d$, i.e., they are d -dimensional vectors with integer components. The increments $X_1, X_2, \dots$ are unit vectors with one component of absolute value 1. Note that $X_j \cdot X_k = 1$ if $j = k$, and if $j \neq k$, then $X_j \cdot X_k$ equals 1 with probability $\frac{1}{2d}$; equals -1 with probability $\frac{1}{2d}$; and otherwise equals zero. In particular, $\mathbb{E}[X_j \cdot X_j] = 1$ and $\mathbb{E}[X_j \cdot X_k] = 0$ if $j \neq k$.

Example 3.5. Based on the definition above, we can create a d -dimensional random walk model using Python, which is shown in *Appendix A*.

The study of higher-dimensional random walks is a natural extension of the one-dimensional case and offers insights into the behaviour of random walks in more complex spaces. One might wonder if the recurrence properties observed in one-dimensional random walks still hold in higher dimensions. To address this question, let us first consider the expected squared distance from the origin after n steps. Suppose $S_0 = 0$. Then $\mathbb{E}[S_0] = 0$, and a similar calculation as in the one-dimensional case gives

$$\mathbb{E}[|S_n|^2] = \mathbb{E}\left[\left(\sum_{j=1}^n X_j\right) \cdot \left(\sum_{k=1}^n X_k\right)\right] = n.$$

If n is odd, the probability that we are at the origin after n steps is zero, assuming $S_0 = 0$. If n is even, we consider the typical distance from the origin of S_n , which is of order $\sqrt{n}$. In d dimensions, the number of lattice points within distance $\sqrt{n}$ grows like $(\sqrt{n})^d$. As a result, the probability that we choose a particular point should decay like a constant times $n^{-\frac{d}{2}}$. The heuristic argument provides an intuitive understanding of the probability of a random walker returning to the origin in higher-dimensional spaces and decaying faster than in the one-dimensional case.

Let us consider the behaviour of simple random walks qualitatively, as the combinatorial arguments required for a rigorous justification are rather more intricate than those encountered in one-dimensional situations. Consider a random walk in which we take $2n$ steps; we anticipate that approximately $\frac{2n}{d}$ of these steps will be taken in each of the d possible directions. For instance, when $d = 2$, we would expect roughly n horizontal and n vertical steps. To find ourselves back at the origin, it is essential

to take an even number of steps in each of the d -directions. The probability of such an occurrence is $2^{-(d-1)}$. Given that each of these numbers is even, the probability that each component is at the origin is tantamount to the probability that a one-dimensional walk is at the origin at time $\frac{2n}{d}$.³ Employing this intuition, we arrive at the asymptotic expression

$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}},$$

where

$$C_d = \frac{d^{\frac{d}{2}}}{\pi^{\frac{d}{2}} 2^{d-1}}.$$

The expression for C_d can be found through a more rigorous analysis of the probabilities associated with the random walk in higher dimensions, specifically by leveraging Fourier analysis techniques on the lattice $\mathbb{Z}^d$. However, the specific value of C_d does not hold great significance for our purposes, so we simply state it here without proof. It is crucial to note that the exponent of n is $\frac{d}{2}$.

Consider the expected number of returns to the origin. Let V represent the number of visits to the origin. Mirroring the approach used for the $d = 1$ case, we obtain

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0).$$

Furthermore, we can express the expected number of visits as

$$\mathbb{E}[V] = \frac{1}{1 - q},$$

where $q = q_d$ denotes the probability that the d -dimensional walk returns to the origin. Recalling that

$$\mathbb{P}(S_{2n} = 0) \sim \frac{C_d}{n^{\frac{d}{2}}},$$

we can now write the expected number of visits as

$$\mathbb{E}[V] = \sum_{n=0}^{\infty} \mathbb{P}(S_{2n} = 0) = \begin{cases} < \infty & \text{if } d \geq 3 \\ = \infty & \text{if } d = 2 \end{cases}.$$

This result uncovers a remarkable aspect of random walks: the expected number of returns to the origin is finite for dimensions greater than or equal to three, while infinite for two-dimensional walks, which is summarised in the following theorem.

Theorem 3.6. *Suppose S_n is a simple random walk in $\mathbb{Z}^d$ with $S_0 = 0$. If $d = 1, 2$, the random walk is recurrent, meaning that with probability one, it returns to the origin infinitely often. If $d \geq 3$, the random walk is transient, signifying that with probability one, it returns to the origin only finitely often.*

Now, let us discuss a lemma that probabilists frequently use. Though it is quite elementary, it has a name. Mathematicians often assign names to commonly used lemmas, even simple ones, as it facilitates easy reference and communication.

³More precisely, an even integer exceedingly close to $\frac{2n}{d}$.

Lemma 3.7 (Borel-Cantelli Lemma). *Suppose $E_1, E_2, \dots$ is a collection of events such that*

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty.$$

Then, with probability one, at most finitely many of the events occur. In other words, the probability that infinitely many events occur is zero.

Proof. Let A represent the event that infinitely many of $E_1, E_2, \dots$ occur. For each integer N , we have $A \subset A_N$, where A_N is the event that at least one of the events $E_N, E_{N+1}, \dots$ occurs. Consequently, we can derive the following inequality

$$\mathbb{P}(A) \leq \mathbb{P}(A_N) = \mathbb{P}\left(\bigcup_{n=N}^{\infty} E_n\right) \leq \sum_{n=N}^{\infty} \mathbb{P}(E_n).$$

However, the condition $\sum \mathbb{P}(E_n) < \infty$ implies that

$$\lim_{N \rightarrow \infty} \sum_{n=N}^{\infty} \mathbb{P}(E_n) = 0.$$

Thus, we conclude that $\mathbb{P}(A) = 0$. □

The *Borel–Cantelli lemma* is an essential tool for understanding the occurrence and frequency of events in probability theory. It helps determine whether a sequence of events will occur infinitely often or only finitely often based on the convergence or divergence of the sum of their probabilities. To better grasp its application, let us consider an example involving a simple random walk.

Example 3.8. Consider the simple random walk in $\mathbb{Z}^d$, where $d \geq 3$, and define E_n as the event that $S_n = 0$. Drawing upon the estimates from our previous discussion, we observe that

$$\sum_{n=1}^{\infty} \mathbb{P}(E_n) < \infty.$$

According to *Lemma 3.7*, with probability one only finitely many of the events E_n occur. Consequently, the random walk visits the origin only finitely often with probability one, in agreement with *Theorem 3.6*.

Chapter 4

Solving Boundary Value Problems

Random walks have found their way into numerous scientific disciplines, consistently exhibiting their versatility and adaptability. In this section, we will explore one significant application of random walks: resolving boundary value problems for harmonic functions, specifically the *Dirichlet problem*. Our journey will commence with the construction of a 1-dimensional random walk model, and then gradually extend it to higher-dimensional cases. Along the way, we will delve into the associated properties of these models, establish the discrete Dirichlet condition as a continuous case analogue, and ultimately employ random walks to solve the Dirichlet problem.

4.1 One Dimension: Gambler's Ruin

Our primary objective in this section is to demonstrate that boundary value problems possess a unique solution in the one-dimensional case. To achieve this, we shall construct a 1-dimensional random walk model—popularly known in statistics as the *gambler's ruin*—and exhibit its unique solution. Let us embark on this quest by introducing some essential definitions.

Definition 4.1. Suppose N is a positive integer and a random walker starts at $x \in \{0, 1, \dots, N\}$. Let S_n denote the position of the walker at time n . The walker stops when it reaches either 0 or N , and we define the *stopping time* as

$$T = \min\{n \mid S_n = 0 \text{ or } S_n = N\}.$$

Then, we define the function $F : \{0, 1, \dots, N\} \rightarrow [0, 1]$ by

$$F(x) = \mathbb{P}(S_T = N \mid S_0 = x),$$

which represents the probability that a random walker starts at position x and ultimately reaches the boundary point N .

We presuppose that the stopping time is finite, a key assumption which we shall substantiate in *Section 4.4*. Indeed, the function $F(x)$ would be rendered undefined if S_T were left undetermined. This connection between the stopping time and the endpoint is indispensable for harnessing the power of random walks to tackle boundary value problems.

Example 4.2 (Gambler's Ruin). We can interpret S_n as the number of chips currently held by a gambler engaged in a fair game where they stand to win or lose one chip at each time duration. The

player begins with x chips and persists in playing until they either amass N chips or exhaust their funds. The probability that the gambler evades bankruptcy before attaining N is given by $F(x)$. Evidently, $F(0) = 0$ and $F(N) = 1$. Suppose $0 < x < N$. After the first game, the gambler has either $x - 1$ or $x + 1$ chips, and both of these outcomes bear equal likelihood. Consequently,

$$F(x) = \frac{1}{2}F(x+1) + \frac{1}{2}F(x-1), \quad (4.1)$$

where $x = 1, \dots, N-1$. One function F that satisfies (4.1) with the *boundary conditions* $F(0) = 0$, $F(N) = 1$ is the linear function

$$F(x) = \frac{x}{N}.$$

Remarkably, this is the only solution, as we shall now elucidate.

Theorem 4.3. *Suppose a, b are real numbers and N is a positive integer. Then the only function $F : \{0, \dots, N\} \rightarrow \mathbb{R}$ satisfying (4.1) with the boundary conditions $F(0) = a$ and $F(N) = b$ is the linear function*

$$\tilde{F}(x) = a + \frac{x(b-a)}{N}.$$

Proof. It is clear that $\tilde{F}$ fulfils the conditions. We have to establish the uniqueness, that $\tilde{F}$ is the *only* such function. We will proceed to demonstrate this in 3 alternative ways.

Method 1. Let us consider a linear function $\tilde{F}$ satisfying the boundary conditions, with $\tilde{F}(0) = a$ and $\tilde{F}(N) = b$. Suppose F is another solution to the boundary value problem described by (4.1) that is distinct from $\tilde{F}$. For each $0 < x < N$, we have

$$F(x) \leq \max\{F(x-1), F(x+1)\}.$$

This inequality implies that $F(x)$ cannot possess local maxima within the interior of the domain $(0, N)$. Consequently, the only possible locations for the maximum of the function F must be at the boundary points $x = 0$ and $x = N$. By analogous reasoning, we deduce that the minimum of F is also achieved on the boundary $\{0, N\}$.

Suppose $F(0) = a$ and $F(N) = b$. We define a new function $F' = F - \tilde{F}$. By construction, F' satisfies (4.1) with $F'(0) = 0$ and $F'(N) = 0$. From our analysis in the previous case, F' must be identically zero. Thus, we obtain the desired result $F = \tilde{F}$, which is the unique solution.

Method 2. Consider the set $\mathcal{V}$ of all functions $F : \{0, \dots, N\} \rightarrow \mathbb{R}$ that satisfy (4.1). In this exploration, we will discover that $\mathcal{V}$ is indeed a vector space. We claim that this vector space has dimension two, which will suffice to find a basis for the vector space with two dimensions. To construct the basis for $\mathcal{V}$, let f_1 be the function that satisfies (4.1) with the initial conditions $f_1(0) = 0$ and $f_1(1) = 1$. Replacing x with $x - 1$ in (4.1), we obtain

$$f_1(x-1) = \frac{1}{2}f_1(x) + \frac{1}{2}f_1(x-2).$$

Consequently, we can define f_1 for $x > 1$ as

$$f_1(x) = 2f_1(x-1) - f_1(x-2)$$

for $x > 1$. We are now tasked with verifying that f_1 is the unique solution to (4.1) satisfying the given

initial conditions. By the recurrence relation, we have

$$f_1(x+1) = 2f_1(x) - f_1(x-1).$$

Rearranging the terms, we get

$$f_1(x) = \frac{1}{2}f_1(x+1) + \frac{1}{2}f_1(x-1).$$

Thus, f_1 satisfies (4.1) for all $x \in \{1, \dots, N-1\}$. Let g_1 be another solution to (4.1) that satisfies the same initial conditions as f_1 , i.e., $g_1(0) = 0$ and $g_1(1) = 1$. Then, for all $x \in \{1, \dots, N-1\}$, we have

$$g_1(x) = \frac{1}{2}g_1(x+1) + \frac{1}{2}g_1(x-1).$$

Similarly, we define $h_1 = f_1 - g_1$, and for all $x \in \{1, \dots, N-1\}$, we have

$$h_1(x) = \frac{1}{2}h_1(x+1) + \frac{1}{2}h_1(x-1).$$

Recalling insights from the previous method, we deduce that the only function satisfying these conditions is the identically zero function. Therefore, $h_1 \equiv 0$, which implies $f_1 \equiv g_1$. This demonstrates that f_1 is indeed the unique solution to (4.1) satisfying the given initial conditions.

Therefore, we define another function f_2 in a similar fashion with the initial conditions $f_2(0) = 1$ and $f_2(1) = 0$. We assert that the unique solution to (4.1) that satisfies $F(0) = c_2$ and $F(1) = c_1$ is given by the linear combination of f_1 and f_2 , i.e., $F = c_1f_1 + c_2f_2$. Observe that the set of functions of the form $\tilde{F}$ constitutes a two-dimensional subspace of $\mathcal{V}$ as a and b vary, and thus it must encompass all of $\mathcal{V}$.

Method 3. Suppose F is a solution to (4.1). Let S_n be the random walk starting at x . We claim that for all n , $\mathbb{E}[F(S_{n \wedge T})] = F(x)$. To substantiate this claim, we will utilise the technique of induction.

For the base case, when $n = 0$, we have $F(S_0) = F(x)$ since $S_0 = x$. Then, it follows that $\mathbb{E}[F(S_0)] = F(x)$. For the inductive step, we employ a rule for expectation in terms of conditional expectations:

$$\mathbb{E}[F(S_{(n+1) \wedge T})] = \sum_{y=0}^N \mathbb{P}(S_{n \wedge T} = y) \mathbb{E}[F(S_{(n+1) \wedge T}) \mid S_{n \wedge T} = y].$$

If $y = 0$ or $y = N$ and $S_{n \wedge T} = y$, then $S_{(n+1) \wedge T} = y$ and hence

$$\mathbb{E}[F(S_{(n+1) \wedge T}) \mid S_{n \wedge T} = y] = F(y).$$

If $0 < y < N$ and $S_{n \wedge T} = y$, then

$$\mathbb{E}[F(S_{(n+1) \wedge T}) \mid S_{n \wedge T} = y] = \frac{1}{2}F(y+1) + \frac{1}{2}F(y-1) = F(y).$$

Therefore, we find that

$$\mathbb{E}[F(S_{(n+1) \wedge T})] = \sum_{y=0}^N \mathbb{P}(S_{n \wedge T} = y) F(y) = \mathbb{E}[F(S_{n \wedge T})] = F(x),$$

with the last equality holding by the inductive hypothesis. Consequently, we obtain

$$\begin{aligned}
F(x) &= \lim_{n \rightarrow \infty} \mathbb{E}[F(S_{n \wedge T})] \\
&= \lim_{n \rightarrow \infty} \sum_{y=0}^N \mathbb{P}(S_{n \wedge T} = y) F(y) \\
&= \mathbb{P}(S_T = 0) F(0) + \mathbb{P}(S_T = N) F(N) = [1 - \mathbb{P}(S_T = N)] F(0) + \mathbb{P}(S_T = N) F(N).
\end{aligned}$$

Considering the case $F(0) = 0$ and $F(N) = 1$, we obtain

$$\mathbb{P}(S_T = N \mid S_0 = x) = \frac{x}{N}.$$

For more general boundary conditions, we deduce that

$$F(x) = F(0) + \frac{x}{N} [F(N) - F(0)].$$

A noteworthy benefit of this approach is that one need not have previously predicted the linear functions to be the solutions. All such solutions are generated by the proof. $\square$

4.2 Higher Dimensions

Now it becomes appropriate to extend our understanding from previous sections to the more encompassing d -dimensional case. To define a function in this context, it is necessary to assign a value to each point in $\mathbb{Z}^d$. Although in the vast expanse of theoretical possibilities, we could attribute any meaning to these assigned values and delve into a myriad of cases, it is within the confines of random walk that only certain aspects pique our interest and warrant further exploration (e.g., interpreting values as temperature or density). We shall first delineate such regions of interest, followed by a discussion on the *mean value property*.

Definition 4.4. Transcending the simplicity of the interval $\{1, \dots, N\}$, we introduce an arbitrary finite subset A of $\mathbb{Z}^d$. The (*outer*) *boundary* of A emerges from this d -dimensional lattice, defined by

$$\partial A = \left\{ z \in \mathbb{Z}^d \setminus A \mid \text{dist}(z, A) = 1 \right\}.$$

With this in place, we let $\bar{A} = A \cup \partial A$ denote the *closure* of A , encapsulating both the interior and the boundary. In this context, the set A is also referred to as the *interior* of $\bar{A}$.

Example 4.5. We can imagine the two-dimensional case with the help of *Figure 4.1*. For the three-dimensional case, it is difficult to visualise, so we use the program in *Appendix A*.

Remark. Drawing inspiration from the realm of continuous spaces, let A be an open set, with ∂A representing its topological boundary and $\bar{A} = A \cup \partial A$ denoting its topological closure. In the discrete setting of $\mathbb{Z}^d$, the boundary in Definition 4.4 is instead defined using lattice distance one.

Definition 4.6. Let A be a subset of $\mathbb{Z}^d$. A function $F : \bar{A} \rightarrow \mathbb{R}$ is said to possess the *mean value property* if for all $x \in A$,

$$F(x) = \frac{1}{2d} \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} F(y).$$

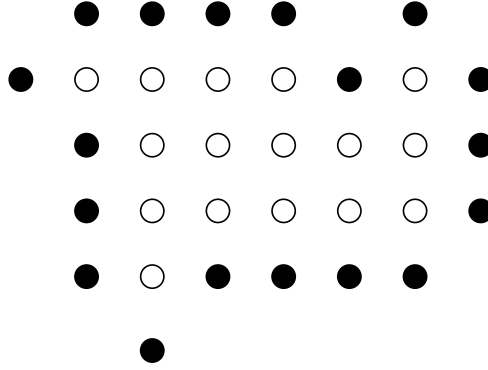


Figure 4.1: The white dots are A and the black dots are ∂A .

To better understand the behaviour of such functions, we define a linear operator $\mathcal{L}$ acting upon a function $F : \mathbb{Z}^d \rightarrow \mathbb{R}$ by

$$\mathcal{L}F(x) = \frac{1}{2d} \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} (F(y) - F(x)).$$

This operator, known as the *(discrete) Laplacian*, plays a pivotal role in our investigation. A function F is deemed *(discrete) harmonic* at a point x if $\mathcal{L}F(x) = 0$. The boundary value problem that emerges from this scenario is sometimes referred to as the *Dirichlet problem* for harmonic functions.

Example 4.7. Let S_n denote a simple random walk in $\mathbb{Z}^d$. By casting our gaze upon the interplay between the Laplacian and the random walk, we can express

$$\mathcal{L}F(x) = \mathbb{E}[F(S_1) - F(S_0) \mid S_0 = x].$$

Remark. The term *linear operator* frequently appears in the context of a linear function acting on a domain comprising a space of functions. In our current exploration, the domain consists of the space of functions on the finite set A , which is isomorphic to $\mathbb{R}^K$, where $K = \#(A)$. In this situation, a linear operator is analogous to a linear transformation from linear algebra. We may envision $\mathbf{Q}$ and $\mathcal{L}$ as $K \times K$ matrices. We can express

$$\mathbf{Q} = [Q(x, y)]_{x, y \in A},$$

where $Q(x, y) = \frac{1}{2d}$ if $|x - y| = 1$, and $Q(x, y) = 0$ otherwise. To delve further into this structure, we define $Q_n(x, y)$ by

$$\mathbf{Q}^n = [Q_n(x, y)].$$

Then the entry $Q_n(x, y)$ unveils the probability that the random walk commences at point x , finds itself at site y at time n , and has not ventured beyond the confines of set A up to time n .

4.3 Dirichlet Problem for Harmonic Functions

The *Dirichlet problem* is a classical and essential question in the realm of boundary value problems, which aims to find a harmonic function within a bounded region that coincides with a given function defined on the boundary. Stated more formally, given a set $A \subset \mathbb{Z}^d$ and a function $F : \partial A \rightarrow \mathbb{R}$, we

seek an extension $\tilde{F} : \bar{A} \rightarrow \mathbb{R}$ of F such that $\tilde{F}$ is harmonic in A . In other words, we want to satisfy the Laplace equation

$$\mathcal{L}F(x) = 0 \tag{4.2}$$

for all $x \in A$. In the one-dimensional case, the Dirichlet problem is considerably simpler. We have a set $A = \{1, \dots, N-1\}$, and the Laplacian operator $\mathcal{L}$ reduces to the second difference operator

$$\mathcal{L}F(x) = \frac{1}{2}(F(x-1) - 2F(x) + F(x+1)).$$

Our task is to find a harmonic function $\tilde{F}(x)$ satisfying $\mathcal{L}F(x) = 0$ for all $x \in A$. The constant function emerges as the most straightforward solution in this one-dimensional case. By setting $\tilde{F}(x) = C$ for some constant C , we can verify that it indeed satisfies the Laplace equation for all $x \in A$ since

$$\mathcal{L}F(x) = \frac{1}{2}(C - 2C + C) = 0.$$

Thus, the constant function $\tilde{F}(x) = C$ is a harmonic function that satisfies the Laplace equation in the one-dimensional case.

In higher dimensions, despite its complexity, we can still generalise the proof for the case of $d = 1$ in a natural way to accommodate $d > 1$. Embarking on this extension, let us first consider a random walk S_n in $A \subset \mathbb{Z}^d$. Just as in the one-dimensional case, we define the *stopping time* as

$$T_A = \min\{n \geq 0 \mid S_n \notin A\},$$

representing the moment when the random walker first reaches the boundary of the set A . In the one-dimensional case, we assumed the stopping time to be finite. Recall that in *Section 3.4*, we derived an important property that the probability of occupying a particular point should decay like a constant times $n^{-\frac{d}{2}}$. For finite A , the random walker leaves A almost surely, so $T_A < \infty$ with probability one.

Theorem 4.8. *If $A \subset \mathbb{Z}^d$ is finite, then for every $F : \partial A \rightarrow \mathbb{R}$, there is a unique extension of F to $\bar{A}$ that satisfies (4.2). It is given by*

$$\tilde{F}(x) = \mathbb{E}[F(S_{T_A}) \mid S_0 = x] = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y \mid S_0 = x)F(y).$$

Proof. We can verify directly that $\tilde{F}$ is a solution to the Dirichlet problem. The real challenge lies in demonstrating the uniqueness of this solution. Suppose F is harmonic on A , $S_0 = x \in \bar{A}$, and let

$$M_n = F(S_{n \wedge T_A}),$$

where $n \wedge T_A$ means the minimum of n and T_A . Then (4.2) can be rewritten as

$$\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = F(S_{n \wedge T_A}) = M_n.$$

A process that satisfies $\mathbb{E}[M_{n+1} \mid S_0, \dots, S_n] = M_n$ is called a *martingale* (with respect to the random walk). It becomes evident that $F(S_{n \wedge T_A})$ being a martingale is essentially equivalent to F being harmonic on A .

It is straightforward to verify that martingales satisfy $\mathbb{E}[M_n] = \mathbb{E}[M_0]$, and hence, if $S_0 = x$, then

$$\sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \mathbb{E}[M_n] = M_0 = F(x).$$

Taking limits, we obtain

$$F(x) = \lim_{n \rightarrow \infty} \sum_{y \in \bar{A}} \mathbb{P}(S_{n \wedge T_A} = y) F(y) = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y) F(y).$$

This completes the proof of the theorem, establishing the uniqueness of the solution to the Dirichlet problem in the context of higher dimensions. $\square$

We can indeed interchange the limit and the sum in our previous discussion, as it is a finite sum. However, if we allow A to be an infinite set, the uniqueness of the solution is no longer guaranteed. To illustrate this point, let us consider a one-dimensional example with $A = \{1, 2, 3, \dots\}$ and $\partial A = \{0\}$. In this case, for every $c \in \mathbb{R}$, the function $F(x) = cx$ is harmonic in A with a boundary value of 0 at the origin.

Suppose F is harmonic on A with $F(0) = 0$, and let S_n be a simple random walk starting at a positive integer x . As before, we let $T = \min\{n \geq 0 \mid S_n = 0\}$ and $M_n = F(S_{n \wedge T})$. The same argument shows that M_n is a martingale and

$$F(x) = \mathbb{E}[M_n] = \sum_{y=0}^{\infty} F(y) \mathbb{P}(S_{n \wedge T} = y).$$

We assumed that with probability one, $T < \infty$. This result implies that $\mathbb{P}(S_{n \wedge T} = 0)$ tends to 1, which can be expressed as

$$\lim_{n \rightarrow \infty} \sum_{y > 0} \mathbb{P}(S_{n \wedge T} = y) = 0.$$

However, if F is unbounded, we cannot conclude from this that

$$\lim_{n \rightarrow \infty} \sum_{y > 0} F(y) \mathbb{P}(S_{n \wedge T} = y) = 0.$$

Nonetheless, our exploration reveals that there is only one bounded function that is harmonic on A with a given boundary value at 0.

Theorem 4.9. *Suppose A is a proper subset of $\mathbb{Z}^d$ such that, for all $x \in A$,*

$$\lim_{n \rightarrow \infty} \mathbb{P}(T_A > n \mid S_0 = x) = 0.$$

Suppose $F : \partial A \rightarrow \mathbb{R}$ is a bounded function. Then there is a unique bounded extension of F to $\bar{A}$ that satisfies (4.2). It is given by

$$\tilde{F}(x) = \mathbb{E}[F(S_{T_A}) \mid S_0 = x] = \sum_{y \in \partial A} \mathbb{P}(S_{T_A} = y \mid S_0 = x) F(y).$$

In conclusion, we have discussed the Dirichlet problem for harmonic functions and demonstrated that the harmonic function satisfying the Laplace equation is unique for finite sets A , and for infinite

sets that satisfy the exit-time condition in *Theorem 4.9*, provided that the boundary function is bounded.

4.4 Expected Time to Escape

In this section, we will prove a lemma that we previously skipped, which is related to the stopping time of a one-dimensional random walker.

Lemma 4.10. *For any one-dimensional random walker, we can assert with certainty (probability one) that $T < \infty$, implying that the walker will eventually reach either 0 or N and subsequently stop.*

Proof. Consider a one-dimensional random walk, denoted by S_n , commencing at a point $x \in \{0, \dots, N\}$. Let T represent the first instance in time when the random walker reaches either of the boundary points, $\{0, N\}$. Our goal is to investigate the expected time to hit one of these boundaries, defined as

$$e(x) = \mathbb{E}[T \mid S_0 = x].$$

It is evident that $e(0) = e(N) = 0$. Now let us assume $x \in \{0, \dots, N-1\}$. At this point, the random walker takes a single step, which could lead to either $x-1$ or $x+1$. Based on this observation, we derive the relation

$$e(x) = 1 + \frac{1}{2} [e(x+1) + e(x-1)].$$

Consequently, e satisfies the following system of equations

$$\begin{cases} e(0) = e(N) = 0 \\ \mathcal{L}e(x) = -1 \end{cases}, \quad (4.3)$$

where $x = 1, \dots, N-1$. Upon further examination, we discover that if $f(x) = x^2$, then $\mathcal{L}f(x) = 1$ for all x . Additionally, the linear function $g(x) = x$ is harmonic, with $\mathcal{L}g \equiv 0$. Utilising this information, we deduce that one solution to (4.3) is

$$e(x) = x(N-x).$$

As we shall demonstrate shortly, this solution is unique. Suppose $\tilde{e}$ is another solution. Then for $x = 1, \dots, N-1$,

$$\mathcal{L}(e - \tilde{e}) = \mathcal{L}e(x) - \mathcal{L}\tilde{e}(x) = 0,$$

implying that $e - \tilde{e}$ is harmonic on $\{1, \dots, N-1\}$. Given that this function also vanishes at 0 and N , we can conclude that $e - \tilde{e} = 0$.

Remarkably, this function is quadratic with respect to x . It reaches its minimum value of 0 at $x = 0$ and $x = N$, and exhibits symmetry to the midpoint, $x = \frac{N}{2}$. When $x \in \{1, \dots, N-1\}$, the function $e(x)$ remains non-negative and finite, since x and $N-x$ are both non-negative and finite. These conditions allow us to conclude that the expected stopping time $e(x)$ is finite for each starting point, which implies that the stopping time T is finite with probability one, ultimately proving the lemma. $\square$

Chapter 5

Conclusion

In this report, we embarked on an investigation of random walks and their applications in boundary value problems. We began our exploration with a one-dimensional random walk, whose probability distribution illustrates the *Central Limit Theorem*. With a firm foundation established, we ventured into the realm of random walks in higher dimensions. We unveiled a dichotomy in their behaviour: random walkers exhibit recurrent tendencies in one- and two-dimensional scenarios, but transient characteristics emerge in higher-dimensional cases. This is further elucidated by *Lemma 3.7*.

Armed with this newfound understanding, we set our sights on the application of random walks to boundary value problems. In the one-dimensional case, we discovered a unique solution. We then extended our analysis to higher-dimensional scenarios and obtained a unique harmonic extension for finite regions, as well as for infinite regions satisfying the exit-time condition in *Theorem 4.9*. In doing so, we unravelled the Dirichlet problem, a quintessential example of a boundary value problem.

Throughout our report, we focused on the discrete case; however, one can naturally extend the conclusions of this report to the continuous case, which would lead us to Brownian motion. Furthermore, we could delve deeper into the properties of the harmonic function and study the heat equation. Due to time constraints, we had to stop here and leave these captivating topics for future endeavours.

In summary, this report offered a comprehensive exposition of random walks and their connection to boundary value problems. By constructing progressively more complex random walks and delving into their properties, we illuminated the intricate nature of random walk models and boundary value problems, thereby providing a solid foundation for future research in this area.

Appendix A

Sample Codes

Central Limit Theorem for One-dimensional Random Walks

```
import numpy as np
import matplotlib.pyplot as plt

def random_walks(num_walks, num_steps):
    walks = np.random.choice([-1, 1], size=(num_walks, num_steps))
    return walks

def scaled_positions(walks):
    positions = np.sum(walks, axis=1)
    return positions / np.sqrt(walks.shape[1])

num_walks = 10000
num_steps = 500

walks = random_walks(num_walks, num_steps)
scaled_pos = scaled_positions(walks)

plt.hist(scaled_pos, bins=50, density=True, alpha=0.6, color='g')

# Plot the standard normal distribution
x = np.linspace(-4, 4, 100)
y = (1 / np.sqrt(2 * np.pi)) * np.exp(-x**2 / 2)
plt.plot(x, y, 'r-', linewidth=2)

plt.xlabel('Scaled Position')
plt.ylabel('Frequency')
plt.legend(['Normal Distribution', 'Random Walk'])
plt.savefig('Example 1.pdf')
```

This code generates multiple random walks, each consisting of a series of steps. Then, we compute

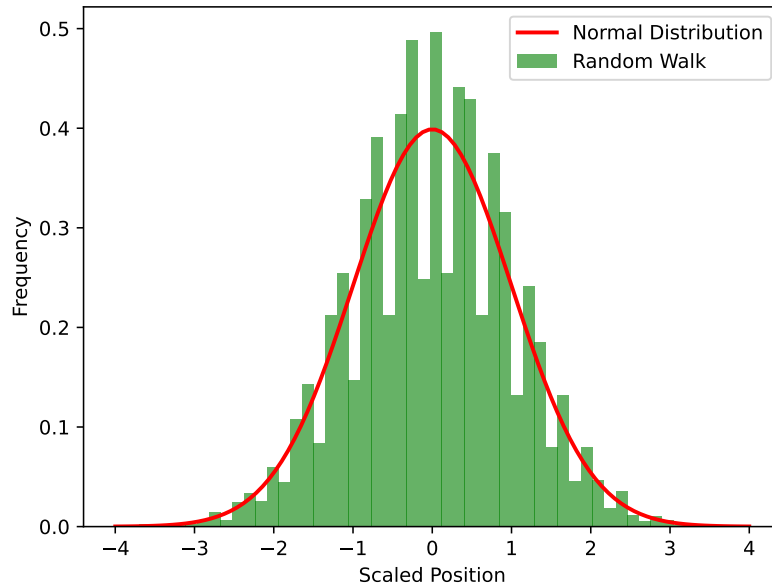


Figure A.1: *Central Limit Theorem* for One-dimensional Random Walks.

the final position for each walk and scale these positions by the square root of the number of steps. Finally, we plot the distribution of these scaled positions. As the number of steps increases, the distribution of the scaled positions converges to a normal distribution with mean 0 and standard deviation 1, which is consistent with the *Central Limit Theorem* for a one-dimensional random walk.

Higher-dimensional Random Walks

```
import numpy as np

def random_walk_d_dimension(n_steps, d):
    """
    Simulate a d-dimensional random walk.

    :param n_steps: int, number of steps in the random walk.
    :param d: int, number of dimensions.
    :return: numpy array, shape (n_steps, d), coordinates of the random walk.
    """
    # At each time, choose one coordinate axis and one direction
    axes = np.random.randint(0, d, size=n_steps)
    directions = np.random.choice([-1, 1], size=n_steps)
    steps = np.zeros((n_steps, d), dtype=int)
    steps[np.arange(n_steps), axes] = directions

    # Calculate the cumulative sum of the steps to get the random walk path
    path = np.cumsum(steps, axis=0)
```

```

    return path

# Set the parameters for the random walk
# n_steps: number of steps in the random walk
# d: number of dimensions (any positive integer greater than 2)
n_steps = 1000
d = 4

# Simulate the random walk using the random_walk_d_dimension function
path = random_walk_d_dimension(n_steps, d)

# The 'path' variable contains the coordinates of the random walk

```

This code provides a general implementation of a d -dimensional random walk in Python using “NumPy”. The main function, `random_walk_d_dimension`, simulates a random walk for a specified number of steps and dimensions. At each time step, it chooses one coordinate axis uniformly and moves one unit in either direction along that axis, as required by the simple random walk model. It then calculates the cumulative sum of the steps to obtain the path.

This version of the code does not include any plotting. To analyse or visualise the random walk in higher dimensions may use dimensional reduction techniques or specialised tools, which are not covered in this report.

Three-dimensional Random Walks with Boundaries

```

import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

def random_walk_3d(steps, boundary):
    # Initialize the starting position at (0, 0, 0)
    x, y, z = 0, 0, 0
    walk_x, walk_y, walk_z = [x], [y], [z]

    # Iterate through the number of steps
    for _ in range(steps):
        # Choose a random axis (0, 1, or 2) and step direction (-1 or 1)
        axis = np.random.choice([0, 1, 2])
        step = np.random.choice([-1, 1])

        # Update the position based on the chosen axis
        if axis == 0:
            x_new, y_new, z_new = x + step, y, z
        elif axis == 1:
            x_new, y_new, z_new = x, y + step, z

```

```

    else:
        x_new, y_new, z_new = x, y, z + step

    # Check if the new position is within the boundaries
    if (-boundary <= x_new <= boundary and
        -boundary <= y_new <= boundary and
        -boundary <= z_new <= boundary):

        # Update the current position
        x, y, z = x_new, y_new, z_new

        # Add the new position to the walk
        walk_x.append(x)
        walk_y.append(y)
        walk_z.append(z)

    return walk_x, walk_y, walk_z

steps = 1000
boundary = 10

walk_x, walk_y, walk_z = random_walk_3d(steps, boundary)

# Create a 3D plot
fig = plt.figure(figsize=(10, 10))
ax = fig.add_subplot(111, projection='3d')

# Plot the 3D random walk
ax.plot(walk_x, walk_y, walk_z, marker='o', markersize=3, linestyle='-')
ax.set_xlim(-boundary-1, boundary+1)
ax.set_ylim(-boundary-1, boundary+1)
ax.set_zlim(-boundary-1, boundary+1)
ax.set_xlabel('X')
ax.set_ylabel('Y')
ax.set_zlabel('Z')
plt.grid(True)
plt.savefig('Example 3.pdf')

```

In this example, we create a 3D random walk with a given boundary. The `random_walk_3d` function takes the number of steps and the boundary as input arguments, and it returns the x , y , and z coordinates of the random walk. The code uses the “NumPy” library to handle the array operations and visualises the random walk using “Matplotlib”.

The random walk starts at position $(0, 0, 0)$ and takes steps in either the positive or negative x , y , and z directions independently. For each step, a random axis is chosen, and a step direction of either

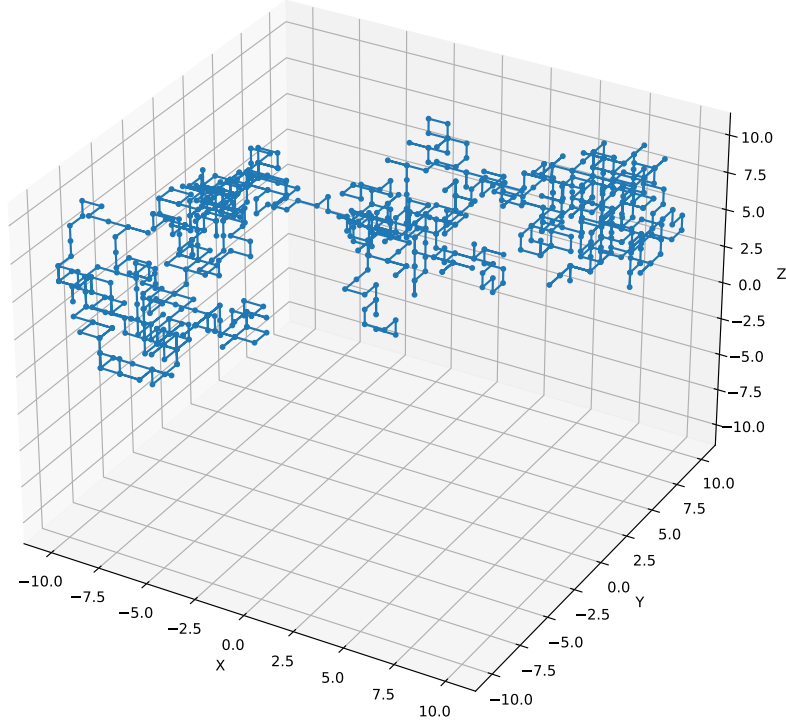


Figure A.2: Three-dimensional Random Walks with Boundaries.

-1 or 1 is selected. If the new position is within the specified boundary, the walker moves to that position, and the position is added to the `walk_x`, `walk_y`, and `walk_z` lists. If the new position is outside the boundary, the walker remains at the current position. Finally, the random walk is plotted in $3D$ with the specified boundary.

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